Ch 2.2: Assessing Model Accuracy

Lecture 3 - CMSE 381

Today's Poll:

Orange Juice?

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Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Weds, Jan 19, 2022

Pulp

No pulp

Announcements

Last time:

- First lab Next friday Jan 28
- Labs are participation only, no upload needed.

Class friday lecture!

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due Fri, Jan 21. Covers:
  - Weds 1/12 lecture
  - Fri 1/14 Lab
  - Weds 1/19 lecture
- No lab this week Yes class!
- Office hours

Covered in this lecture

- Mean Squared Error (regression)
- Train vs Test
- Bias Variance Trade off
- Error rate (classification)
- Bayes Classifier
- K-NN classification

Quick review of notation

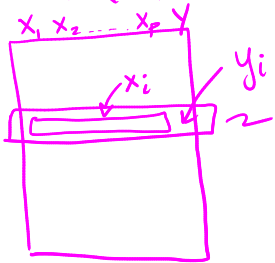
Input variables $X_1, \ldots, X_p$ ← $p = \#$ variables

Output Y

Ground truth $Y = f(X) + \varepsilon$ ← holding all the X_i's

Approximation $\hat{Y} = \hat{f}(X)$

Data:



← data point

(X_1, Y_1)

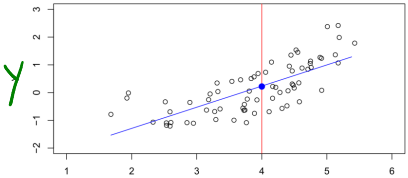
$(X_2 \; Y_2)$

$\vdots$

(X_n, Y_n) ← $\#$ data points

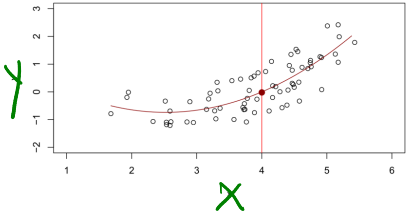
Which is better?

A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



$p = 1$

A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.



No free lunch

No best model for <u>all</u> data sets.

So, we need to <u>measure</u> the quality of any model on a given data set.

Section 1

Mean Squared Error

Mean Squared Error (MSE)

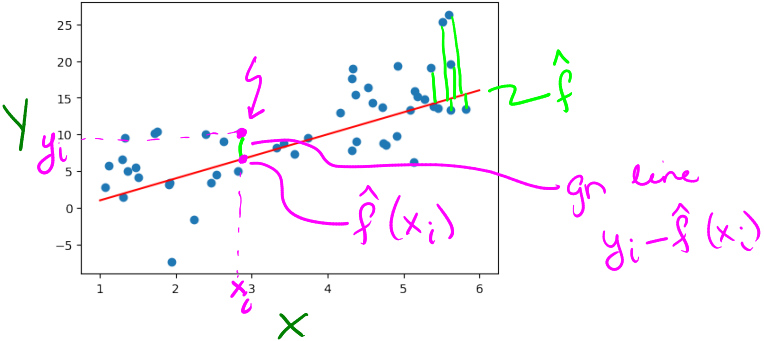
Error in the regression setting

model to be evaluated

(x_i, y_i)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

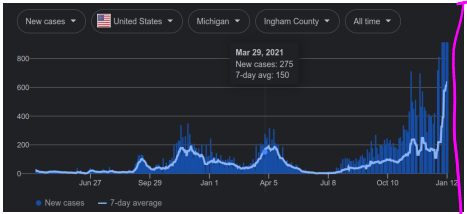
for each data pt



Training MSE

MSE: calculated on the given data, eval. $\hat{f}$

I don't care!



What I want:
MSE of the model on data I haven't seen yet

How good will the model be on tomorrow's decisions?

Train vs test

Training set:
The observations
$\{(x_1, y_1), \cdots, (x_n, y_n)\}$ used to get
the estimate $\hat{f}$

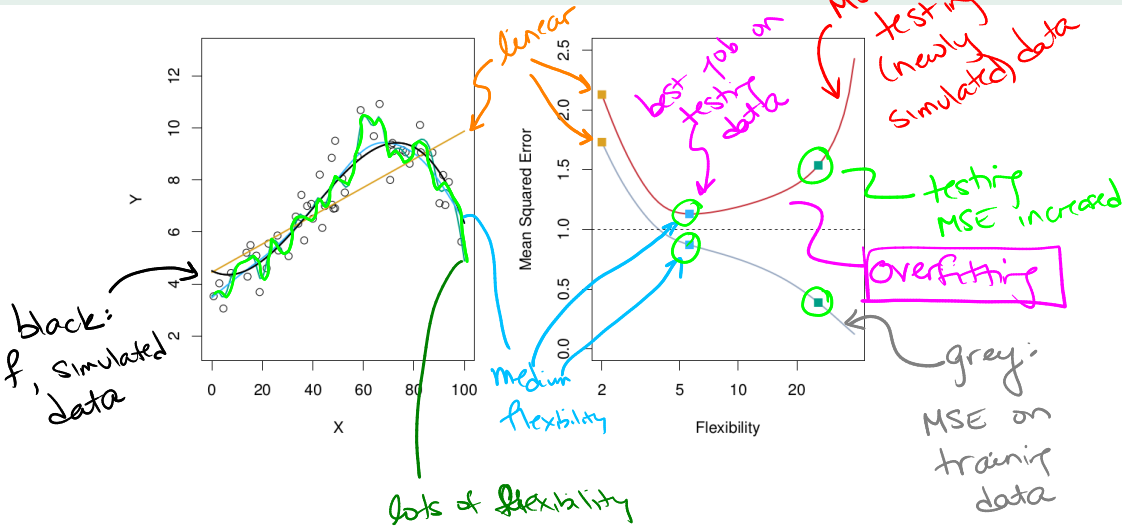
different data

Test set:
The observations
$\{(x'_1, y'_1), \cdots, (x'_{n'}, y'_{n'})\}$ used to
compute the average squared
prediction error

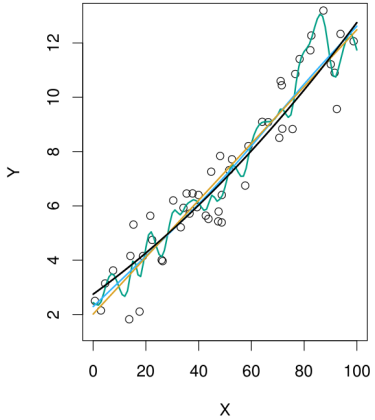
$$\frac{1}{n'} \sum_i (y'_i - \hat{f}(x'_i))^2$$

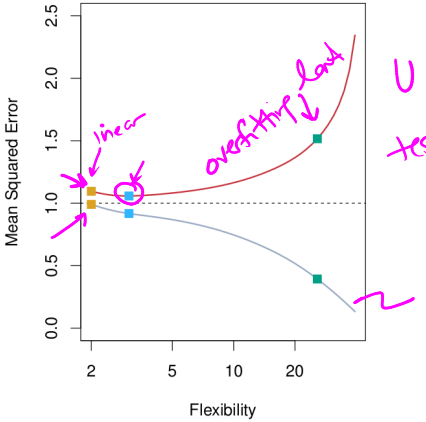
computed on
unseen data!

Why not just get the best model for the training data?



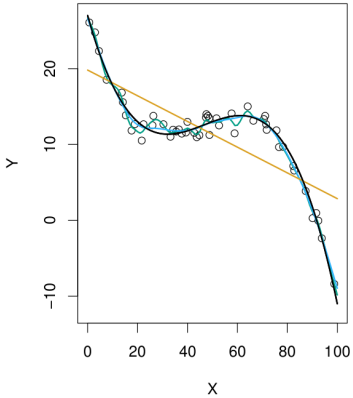
A more linear example

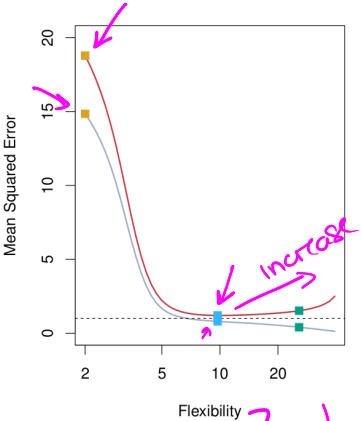




U-shape in test MSE

A more non-linear example





increase in testing MSE

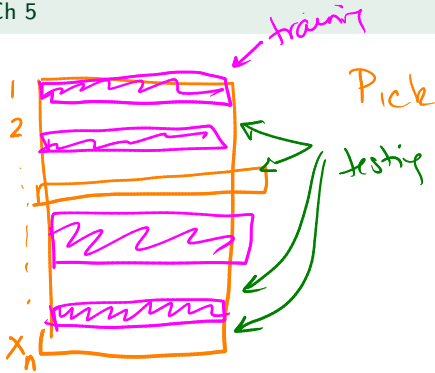
some overfitting

2 degrees of freedom

- Similar U-shape
- More flexibility before increase

A simple solution: Train/test split

More on this in Ch 5



Pick percent $\overbrace{70\%}^{\text{randomly selected}} / 30\%$

$\underbrace{70\%}_{\text{use 70\% of data for training}}$ → used to build $\hat{f}$

$\underbrace{30\%}_{\text{use 30\% of data for testing}}$ → score MSE

Section 2

Bias-Variance Trade-Off

Bias-variance

sampled the data

want both to be low

irreducible error

$$\underbrace{E(y_0 - \hat{f}(x_0))^2}_{} = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

- Expected MSE at x_0

  Avg MSE if repeatedly estimate f w/ newly sampled data sets

- Computed by averaging $E(y_0 - \hat{f}(x_0))^2$ over all values in test

$$E(y_0 - \hat{f}(x_0))^2 \geq \mathrm{Var}(\varepsilon)$$

$$Y = f(X) + \varepsilon$$

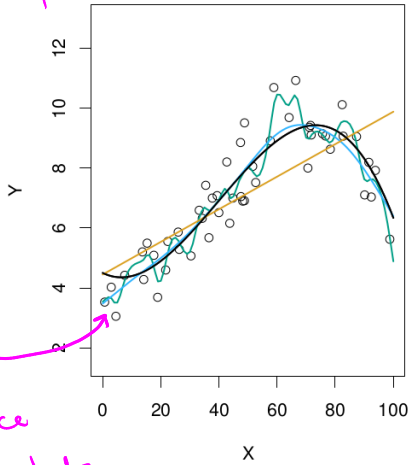
- Proof outside the scope of class

Variance

High Variance: small changes in training data results in large changes in $\hat{f}$

$Var(\hat{f}(x_0))$

Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.



teal

high variance

slightly diff data

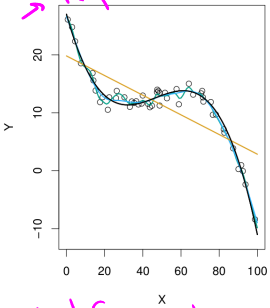
⇒ very different wiggles

Bias

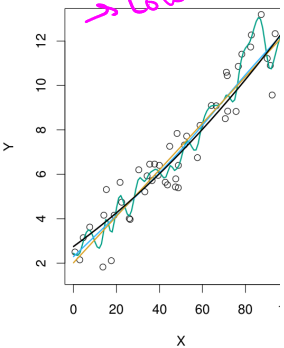
Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.

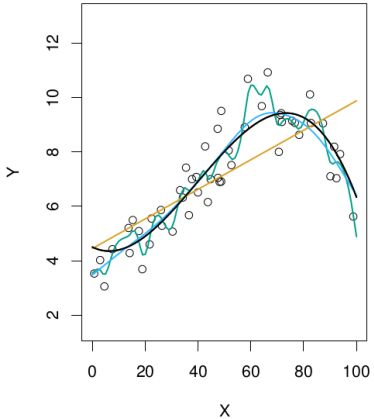
Ex. Linear regression
Likely too simple for any real life problem

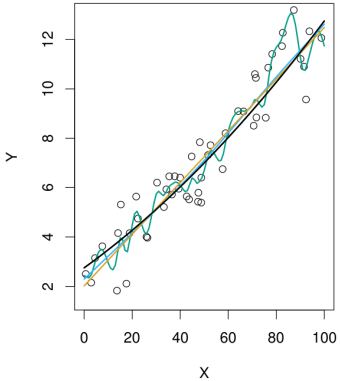
Linear regression
→ High bias

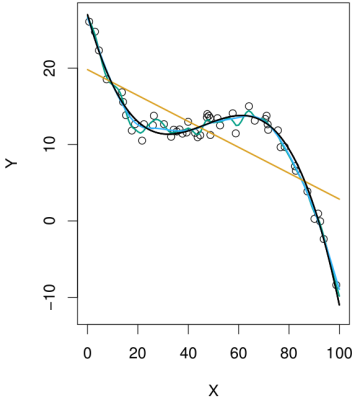


Linear regression
→ Low bias



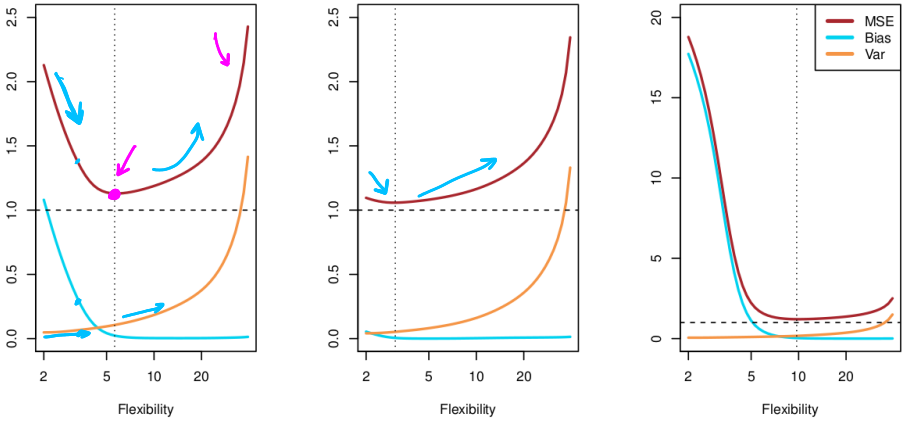






$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

no control



$$\overbrace{E(y_0 - \hat{f}(x_0))^2}^{\text{MSE}} = \overbrace{\mathrm{Var}(\hat{f}(x_0))}^{\text{increasing}} + \overbrace{\left[\mathrm{Bias}(\hat{f}(x_0))\right]^2}^{\text{decreasing}} + \mathrm{Var}(\varepsilon)$$

Bias-variance trade off

- More flexible: variance up, bias down.
- Relative rate of change determines whether MSE goes up or down
- Initially, bias goes down faster, so MSE declines
- Eventually little effect on bias but lots of increased variance, so MSE increases

↳ Results in U-shape of MSE

$$MSE = \frac{1}{n}\sum \left(y_i - \hat{f}(x_i)\right)^2$$

green

red

Section 3

Classification

↳ predict y in set

↳ yes/no

↳ types of drugs

Error rate

- Training data: $\{(x_1, y_1), \cdots, (x_n, y_n)\}$ with y_i qualitative
- Estimate $\hat{y} = \hat{f}(x)$
- Indicator variable

Training error rate:

$$\underbrace{\frac{1}{n}\overbrace{\sum_{i=1}^{n}\underbrace{\mathrm{I}(y_i \neq \hat{y}_i)}}}$$

# of incorrect guesses

percentage of incorrect guesses

Test error rate:

$$\mathrm{Ave}(\mathrm{I}(y_0 \neq \hat{y}_0))$$

correct guess

$$I(y_i \neq \hat{y}_i) = \begin{cases} 1 & \text{if } y_i \neq \hat{y}_i \text{ Wrong} \\ 0 & \text{if } y_i = y_i \text{ Right} \end{cases}$$

Image process

Pic

Pic

Best ever classifier

We can't have nice things

Bayes Classifier:
Give every observation the highest probability class given its predictor variables

$$\Pr(Y = j \mid X = x_0)$$

An example

- Survey students for amount of programming experience, and current GPA
- Try to predict if they will pass CMSE 381.
- If we have a survey of all students that could ever exist, we can determine the probability of failure given combo of those features.

x_1 x_2

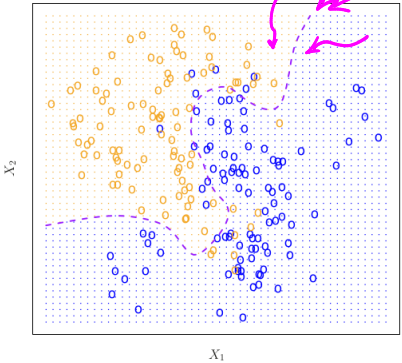
$\rightarrow y$

No single classifier ever perfect since students w/ same info might have diff outcomes

All people ever $(3\text{yrs}, 3.7) \rightarrow$ what % passed

Bayes decision boundary

2-class classifier Blue/Orange



purple line

where $P(Y=j \mid X=x_0) = 50\%$

Bayes error rate

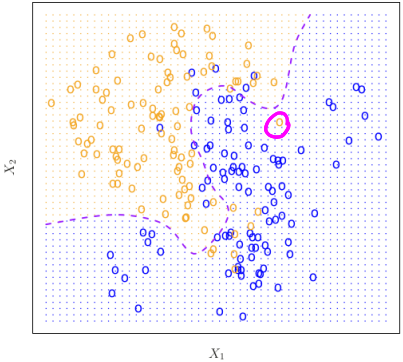
- Error at $X = x_0$

$$\underbrace{1 - \max_j \Pr(Y = j \mid X = x_0)}_{\text{Prob of being wrong}}$$

- Overall Bayes error:

$$1 - E\left(\max_j \Pr(Y = j \mid X = x_0)\right)$$

Irreducible error

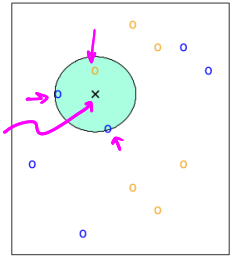


The game

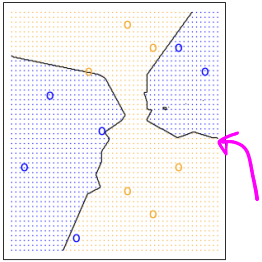
To do as good a job as possible at estimating the Bayes decision body

Classify given observation w/ the highest estimated probability

K-Nearest Neighbors



$K = 3$



decision boundary

- Fix K positive integer
- $N(x)$ = the set of K closest neighbors to x
- Estimate conditional proability

$$\Pr(Y = j \mid X = x_0) = \frac{1}{K} \underbrace{\sum_{i \in N(x_0)} \mathrm{I}(y_i = j)}_{\frac{1}{3} \cdot 2 = \frac{2}{3}}$$

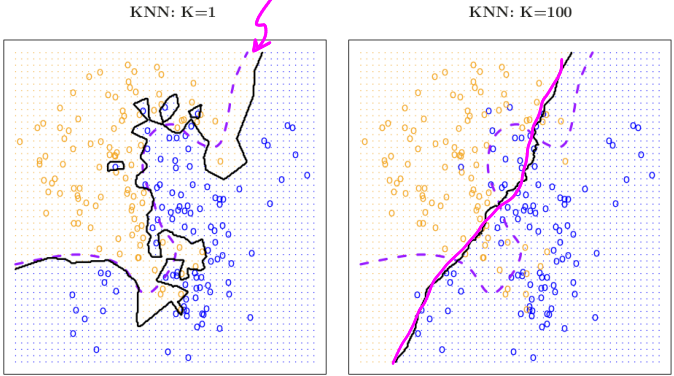
- Pick j with highest value

j = blue $\rightsquigarrow \frac{2}{3}$

j = orange $\frac{1}{3}$

Tradeoff

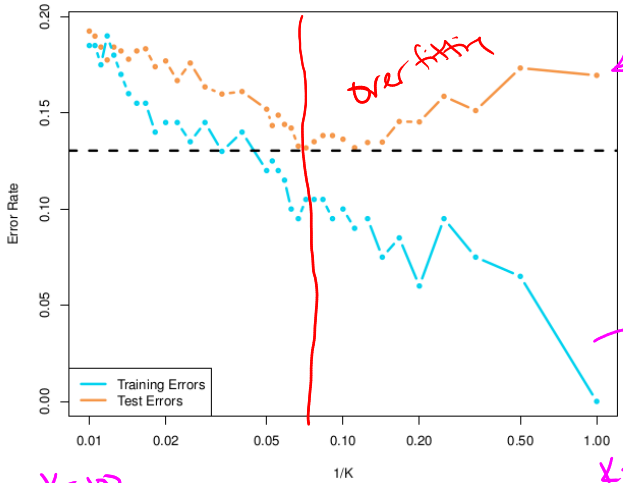
correct Bayes decision boundary



very flexible
overfitting

not flexible

More on tradeoff



overfitting

testing error

U-shape similar to regression setting

training error

K=100 less flexible

K=1 more flexible

Next time

- Friday:
  - Not a lab!
  - 3.1 Linear Regression
  - Homework due midnight on D2L

↳ # 1

⇓

Office hours

MSE

Bias - variance

Overfitting

Classification error